

Hydrochemical characterization and assessment of groundwater in the Bechar-Kenadsa Coal Basin (Southwestern Algeria)

Caratterizzazione e valutazione idrochimica delle acque sotterranee nel bacino carbonifero di Bechar-Kenadsa (Algeria sud occidentale)

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Abstract

In arid and semi-arid zones, groundwater is indispensable but has become extremely limited due to climate change, salinization, overexploitation, and other anthropogenic factors. This study assessed the hydrogeochemistry and quality of groundwater in the Bechar-Kenadsa region of Southwestern Algeria. A total of 37 borehole water samples tapping the aquifer system were collected across the entire study area using a simple random sampling method. These samples were analyzed for physicochemical parameters, as well as major cations and anions, using standard analytical methods. Hydrogeochemical characteristics were assessed using various software tools, including QGIS for cartography, XLSTAT, and diagrams such as Piper, Gibbs, Gaillardet, and Schoeller-Berkaloff. The results showed that the pH ranges from 6.70 to 7.97, while water temperatures vary minimally between 24.5 °C and 24.7 °C. Electrical conductivity ranges between 1993 and 16990 µS/cm. Regarding major ions, Ca²⁺ ranges from 45 to 915 mg/L, Mg²⁺ from 72 to 645 mg/L, and combined sodium and potassium (Na⁺, K⁺) range between 190.2 and 2718 mg/L. For anions, chloride (Cl⁻) ranges from 300 to 4400 mg/L, sulfate (SO₄²⁻) from 11 to 3200 mg/L, bicarbonate (HCO₃⁻) from 64 to 308 mg/L, and nitrate (NO₃⁻) from 2 to 332 mg/L.

The groundwater from this aquifer is used exclusively for agricultural purposes. Overall, these waters are highly mineralized, of poor quality, and not potable. This degraded water quality is attributed to the regional paleogeographic setting of coal formations, the vulnerability of the aquifer, the dissolution of slag heaps, the excessive use of fertilizers, and urban wastewater discharges. Furthermore, effective groundwater protection will rely on wastewater treatment, reduced fertilizer application, the protection of wellhead capture zones, regular water quality monitoring, and sustainable resource management.

Riassunto

Nelle zone aride e semi-aride, le acque sotterranee sono indispensabili ma sono diventate estremamente limitate a causa dei cambiamenti climatici, della salinizzazione, dello sfruttamento eccessivo e di altri fattori antropogenici. Il presente studio ha valutato l'idrogeochimica e la qualità delle acque sotterranee nella regione di Bechar-Kenadsa, nel sud-ovest dell'Algeria. Sono stati raccolti in totale 37 campioni d'acqua da pozzi che attingono al sistema acquifero in tutta l'area di studio, utilizzando un metodo di campionamento casuale semplice. Questi campioni sono stati analizzati per determinare i parametri fisico-chimici, nonché i principali cationi e anioni, utilizzando metodi analitici standard. Le caratteristiche idrogeochimiche sono state valutate utilizzando vari strumenti software, tra cui QGIS per la cartografia, XLSTAT e diagrammi quali quelli di Piper, Gibbs, Gaillardet e Schoeller-Berkaloff. I risultati hanno mostrato che il pH varia da 6,70 a 7,97, mentre la temperatura dell'acqua oscilla in misura minima tra 24,5 °C e 24,7 °C. La conduttività elettrica varia tra 1993 e 16990 µS/cm. Per quanto riguarda gli ioni principali, il Ca²⁺ varia da 45 a 915 mg/L, il Mg²⁺ da 72 a 645 mg/L, mentre il sodio e il potassio combinati (Na⁺, K⁺) variano tra 190,2 e 2718 mg/L. Per quanto riguarda gli anioni, il cloruro (Cl⁻) varia da 300 a 4400 mg/L, il solfato (SO₄²⁻) da 11 a 3200 mg/L, il bicarbonato (HCO₃⁻) da 64 a 308 mg/L e il nitrato (NO₃⁻) da 2 a 332 mg/L. Le acque sotterranee provenienti da questa falda acquifera sono utilizzate esclusivamente per scopi agricoli. Nel complesso, queste acque sono altamente mineralizzate, di scarsa qualità e non potabili. Questo deterioramento della qualità dell'acqua è attribuito al contesto paleogeografico regionale caratterizzato da formazioni carbonifere, alla vulnerabilità della falda acquifera, alla dissoluzione dei cumuli di scorie, all'uso eccessivo di fertilizzanti e agli scarichi delle acque reflue urbane. Inoltre, un'efficace protezione delle acque sotterranee dipenderà dal trattamento delle acque reflue, dalla riduzione dell'uso di fertilizzanti, dalla tutela delle zone di captazione dei pozzi, dal monitoraggio regolare della qualità dell'acqua e dalla gestione sostenibile delle risorse.

Introduction

Groundwater constitutes a strategic and indispensable natural resource for sustaining life. In arid and semi-arid regions, where surface water resources are scarce, it represents the primary source of supply for domestic consumption, irrigation, and industrial needs (Adimalla et al., 2018; Li & Qian, 2018). This strong dependence on groundwater, combined with pollution and increasingly arid and semi-arid climatic conditions, has led to intensified overexploitation of aquifers as well as a progressive deterioration in their quality (Bellaredj & Hamidi 2023; Bellaredj, 2025; Benmarce et al., 2023; Ray et al., 2017; Li et al., 2017; Zahedi, 2017). The chemical composition of groundwater is closely linked to its hydrogeological context, being strongly influenced by interactions between water and the elements encountered along its flow path (Liu et al., 2019). As groundwater circulates through geological formations (Ferlatte et al., 2014; Collin, 2004; Margat, 1998; Castany, 1982; De Marsily, 1981), it undergoes various geochemical processes, such as mineral dissolution, ion exchange, and redox reactions, which progressively modify its chemical composition (Olea-Olea et al., 2020). Indeed, hydrogeochemistry constitutes a fundamental tool for understanding the origin, chemical composition, and flow mechanisms of groundwater. This analytical framework facilitates the identification of water–rock interactions and the underlying geochemical processes that govern the quality and mobility of chemical elements within aquifer systems (Daniele et al., 2013). Furthermore, groundwater can also be affected by contaminants of natural or anthropogenic origin (Akhtar et al., 2021). In arid and semi-arid regions, the salinization of aquifers constitutes a major issue to be addressed (Belkendil et al., 2018; 2021; 2025; Soukaina et al., 2024; Kabour & Chebbah, 2017). The quality of groundwater is influenced by a combination of natural and anthropogenic factors. Among the natural processes, soil composition, evaporation, and precipitation play a predominant role in the variation of the chemistry of these resources (Abhilash-Rani et al., 2020). Moreover, human activities, particularly agriculture, waste management, and the discharge of urban and industrial wastewater, exert a significant influence on groundwater quality (Akhtar et al., 2021). It is therefore essential to accurately characterize the hydrochemistry of groundwater in order to enable effective and sustainable management of water resources in the study area.

The Bechar region, located in southwestern Algeria, is characterized by an arid climate. Its geology comprises several aquifers of varying importance, the study of which remains partial and heterogeneous. The Turonian aquifer is the most exploited and the best studied, belonging to the Cretaceous basin of Bechar (Lachache et al., 2018; 2026; Sadat et al., 2020; Idrotechnico, 1976). The Jurassic aquifer, less extensive, is confined to the Mougheul region and is part of the Atlas domain (Aloui, 2019; Mebarki, 2012; Mekkaoui, 2000; Bourgeois, 1960). The Namurian–Visean aquifer is scarcely exploited due to its low yield (Idrotechnico, 1976). The Eocene aquifer remains poorly understood, and its potential is unclear (Idrotechnico, 1976). Finally, the Westphalian

system is largely unknown (Guerre, 1975; Idrotechnico, 1976), except for a few fragmentary studies (Ouda & Tidjini, 2014; Brahmi, 2015; Kabour & Chebbah, 2017; Sellami & Rabah, 2017). Indeed, the Kenadsa region hosts a complex aquifer system that is still poorly studied. The aquifer formations are composed of Upper Carboniferous (Westphalian) sandstones and are part of the Bechar coal basin (Deleau, 1951; 1952; 1955). This basin was exploited for coal from 1918 to 1975 due to the richness of its carboniferous formations (Benyoucef, 2012; Lemosquet et al., 1976; 1980; 1985; Pareyn, 1961; Levy, 1949; Flamand, 1911; Poirmeur, 1906). Since 1980, this area has been converted into an agricultural zone.

The assessment of groundwater quality within the hydrogeological complexities of the Kenadsa coal basin necessitates a robust integrated framework, as current research often lacks a synthesized approach to distinguishing between natural geochemical evolution and anthropogenic degradation. While statistical methodologies, water quality indices (EWQI), and pollution indices are established tools for assessing the potability and environmental state of water (Bellaredj & Hamidi 2023; Bellaredj, 2025), this study introduces a novel analytical framework by synergistically employing these metrics alongside a comprehensive suite of diagnostic diagrams specifically the Piper, Schoeller-Berkaloff, SAR, Gibbs, and Gaillardet plots to provide a high-resolution diagnostic of the region's hydrogeochemical system. The primary objectives of this research are to classify groundwater according to its ionic composition and rock-weathering mechanisms, evaluate the suitability of resources for both human consumption and irrigation, and isolate the specific natural and anthropogenic drivers such as nitrate concentrations and Total Dissolved Solids (TDS) that control water quality in this vulnerable arid environment. Ultimately, this multifaceted analysis establishes a scientific foundation for sustainable water resource management, adhering to the principle that a comprehensive understanding of hydrochemical dynamics is essential for the effective stewardship of resources facing intensified human and climatic pressures: “study to better understand, better understand to better manage.”

Materials and methods

Study area

The study area is situated within the Kenadsa Oasis in the Bechar Province of southwestern Algeria, encompassing a rectangular corridor approximately 25 km in length and 8 km in width that extends from Kenadsa to Béchar Djedid. Geologically, the region is integrated into the Westphalian coal basin, characterized by a complex stratigraphy of fractured Westphalian sandstones. Hydrogeological evaluations of these formations indicate a permeability $K=10^{-5}$ m/s and a transmissivity $T=10^{-6}$ m²/s (Amri et al., 2023). The presence of numerous resurgences across the landscape suggests that the aquifer roof is located at relatively shallow depths. Regarding water utility and hydrochemistry, the groundwater is utilized exclusively for agricultural purposes due to its poor quality; it

is highly mineralized and classified as non-potable. The study area is geographically demarcated by Cretaceous formations, locally referred to as the 'Première barga,' while its lateral boundaries are defined by the Bechar wadi to the east and the Messaouer wadi to the west. The southern limits are marked by the topographical features of Guelab el Guemmah and Menouna (Fig. 1).

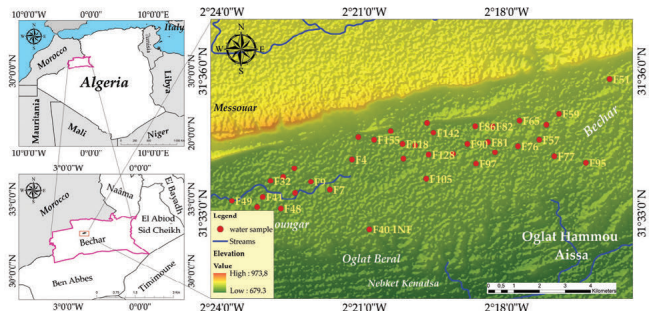


Fig. 1 - Location map of study area with sampling sites in the Bechar-Kenadsa.

Fig. 1 - Carta dell'area di studio indicante i siti di campionamento nell'area di Bechar-Kenadsa.

The coal basin corresponds to the northern flank of the Menouna anticline. At its boundaries, the main geological formations can be briefly identified. The Westphalian basement consists of a thick alternation of sandstones, gypsum-bearing marls, and shales, interrupted by thin beds of bioclastic limestones. This lithological succession contains coal veins, which justifies the designation "Productive Carboniferous" (Deleau, 1951; 1952; 1955). Above this basement lies a secondary and tertiary cover, which rests in a discontinuous and discordant manner over the Carboniferous basement. The most recent formation is the quaternary, limited to wadi alluvium, regs (flat, gravel-covered desert

surfaces formed by wind erosion), and Sebkhha deposits. (Fig. 2).

According to previous studies (Deleau, 1951; 1952; 1955), the Westphalian is subdivided into two main units. The Lower Westphalian (1200 m) consists of approximately 200 m of red gypsum marls, followed by a thick alternation of red gypsum marls and sandstones measuring about 1000 m. The Upper Westphalian (1400 m) begins with gypsum marls and schists containing coal veins, overlain by a marker bed of conglomerates ranging from 80 to 180 m in thickness. This is followed by an alternation of schists and sandstones, with thin limestone beds appearing in the upper part of the sequence, interrupted by a few conglomerate horizons (Fig. 3).

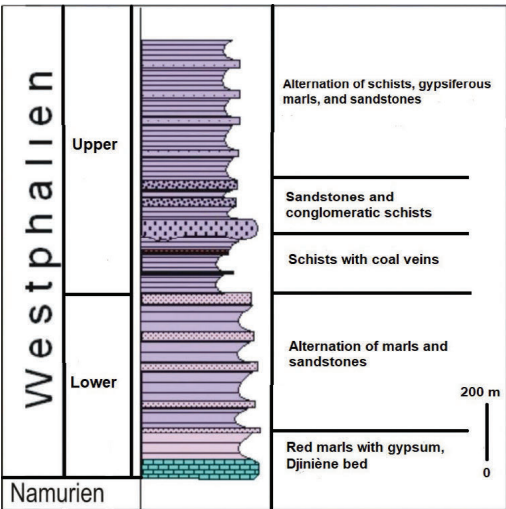


Fig. 3 - Westphalian stratigraphic section of the Bechar-Kenadsa coal basin (Deleau, 1951; 1952; 1955).

Fig. 3 - Sezione stratigrafica del Westfaliano nel Bacino Carbonifero del Bechar - Kenadsa (Deleau, 1951; 1952; 1955).

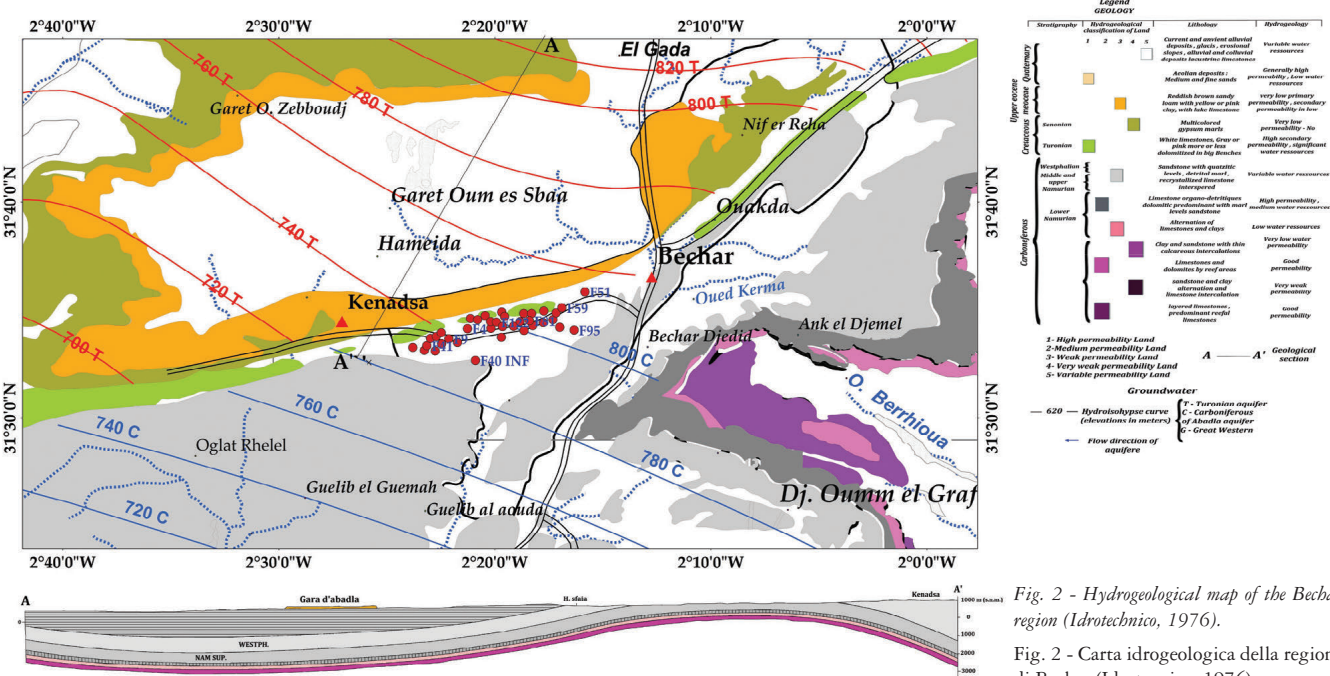


Fig. 2 - Hydrogeological map of the Bechar region (Idrotecnico, 1976).

Fig. 2 - Carta idrogeologica della regione di Bechar (Idrotecnico, 1976).

The Westphalian is folded into an anticline due to tectonic forces, causing the stratigraphic sequence to respond unevenly; marl and clay formations absorb the stresses and exhibit creep and irregular clay behavior. In contrast, competent rocks fracture. These deformations generate faults that act as drains between the different aquifers. The recharge of this system is supplied by direct precipitation on outcrops and by the flows of the Bechar wadi, becoming increasingly rare, while taking into account the years of excess. In general, the region is characterized by low rainfall. Precipitation is irregular and extends from October to March with an annual average of 72 mm. The low temperatures of 4°C were recorded in January and the high value of 40°C in July, with an average of 27°C. The average evaporation is 306 mm, while the evapotranspiration values exceed those of the precipitations implying a drought throughout the year (Lachache et al., 2018; 2026). Desert grasses and palms are the most critical components of the vegetation cover in this region. Climatic data recorded at the Bechar station (Dubief, 1963; ONM, 2018) indicate that the Bechar climate is arid. The groundwater flow is generally oriented toward the southwest at the regional scale. whereas, at the local scale particularly within the Westphalian sandstone aquifer, the flow is predominantly from east to west (Belkendil, 2018; 2021; 2025).

Data and Analysis

In this study, a water sampling campaign was conducted on 37 boreholes over the period from April 15 to June 26, 2023. These structures, which tap the aquifer system, are distributed across the entire study area. Representative boreholes were selected for the analysis of results. In situ measurements of physicochemical parameters were carried out. pH was measured using a WTW-720 pH meter ($\pm 0.1\%$), while electrical conductivity (EC) was measured using a Delta Ohm conductivity meter ($\pm 0.5\%$). The samples were collected in 1 Liter plastic bottles and subsequently analyzed at the Chemistry Laboratory of the National Agency of Hydraulic Resources in Adrar (Algeria). During transport to the laboratory, standard precautions were implemented to preserve sample integrity, the bottles were properly labeled and tightly sealed to prevent any contamination or leakage, and stored in an insulated cooler at low temperature ($= 4^\circ\text{C}$) to limit physicochemical and biological reactions. Transport was carried out as quickly as possible to minimize any alteration of the measured parameters. Finally, the samples were subjected to a series of physicochemical analyses following Rodier's water analysis method (Rodier, 1996; 2009).

The concentration of calcium and magnesium was determined by the complexometric titration with EDTA ($\approx 0.1 - 1 \text{ mg/L}$); chlorides by Argentometric titration ($\approx 0.1 - 1 \text{ mg/L}$); sulfates (SO_4^{2-}) and nitrates (NO_3^-) assessed with a UV-visible PERKIN ELMER-lambda 25 (≈ 1 and 0.1 mg/L respectively); sodium (Na^+) and potassium (K^+) by flame photometer (Model: ELICO CL 378) (precision $\pm 0.5 \text{ ppm}$); The concentrations of HCO_3^- were measured by

Volumetric titration with H_2SO_4 ($\approx 1 \text{ mg/L}$ as CaCO_3). The reliability and internal consistency of the analytical data were verified using the principle of ionic balance. The results indicate that the samples collected during the campaign provide a robust estimation of the chemical composition. For the processing of the results, different software was used; in particular, Cartographic software QGIS (V 3.34.9 Prizren) was used to define the study region and distribute all boreholes in the study area. Diagrammes (V 9.32) software was used to generate the Piper diagram, the Schoeller-Berkaloff diagram and the SAR diagram. The XLSTAT 2016 software was used to generate the correlation table and correlation curves between the different chemical elements. Pearson correlation was employed to evaluate linear relationships between hydrochemical parameters and identify the dominant ions controlling overall mineralization, facilitating an integrated analysis with Principal Component Analysis (PCA). This mathematical data-reduction technique simplified the multidimensional dataset by condensing numerous variables into a limited set of significant components while preserving essential information. By synthesizing these multivariate statistical approaches, the study effectively highlighted parameter interrelationships and characterized the underlying factors governing groundwater quality and the hydrogeochemical integrity of the aquifer system

Hydrogeochemical facies were identified using Piper and Schoeller-Berkaloff diagrams. The Gibbs diagram identifies the dominant factors controlling water chemistry specifically rock-water interaction, evaporation, or precipitation while the Gaillardet diagram refines this diagnostic by pinpointing specific mineralization sources, such as silicate, carbonate, or evaporite weathering. Together, these tools facilitate a rigorous characterization of the natural geochemical processes governing aquifer quality. Groundwater suitability for human consumption was assessed using the Potable Water Quality Index (PWQI), while electrical conductivity (EC) and the sodium adsorption ratio (SAR) were used to evaluate suitability for irrigation

The potable water quality index (PWQI) allows the assessment of the suitability of a given water for human consumption, based on a set of selected physicochemical parameters that are weighted and compared to standard reference values (Diongue et al., 2022). In this study, the value of ten parameters was used to calculate the PWQI, namely: Cl^- , HCO_3^- , NO_3^- , SO_4^{2-} , Ca^{2+} , Mg^{2+} , Na^+ , K^+ , TDS, and pH. The TDS, NO_3^- and Cl^- parameters received a maximum weight of 5 due to their significant impact on water quality (Wagh et al., 2019). The Na^+ and Ca^{2+} ions were assigned weights of 4 and 3, respectively, because of their major role in regulating water quality. The parameters SO_4^{2-} , Mg^{2+} , and pH were assigned weights of 2 and HCO_3^- , K^+ were assigned weights of 1, depending on their influence on water quality. The weights (W_i) in the PWQI are assigned based on the relative importance of water quality parameters for human health and drinking water suitability, considering WHO guidelines, health impacts, hydrochemical significance,

and anthropogenic pollution (Bellaredj & Hamidi 2023; Bellaredj, 2025). The calculation formulas and the water quality classification according to the PWQI are presented in Tables 1 and 2.

Tab. 1 - Parameters and formulas for the PWQI.

Tab. 1 - Parametri e formule per PWQI.

Parameters	Formulas
The relative weight of parameters (W_{ri})	$W_{ri} = \frac{w_i}{\sum_{i=1}^n w_i}$
The quality rating scale (q_i)	$q_i = \frac{C_i}{S_i} \cdot 100$
The water quality sub-index (WQS_i)	$WQS_i = W_{ri} \cdot q_i$
The potable water quality index ($PWQI$)	$PWQI = \sum WQS_i$
With : n : number of parameters, w_i : assigned weight, C_i : concentration of each parameter, S_i : standard value of each parameter	

Tab. 2 - Water class based on PWQI values (Aly et al., 2015).

Tab. 2 - Classificazione delle acque basata sui valori PWQI (Aly et al., 2015).

PWQI values	Water class
< 50	Excellent
50 - 100	Passable
100 - 200	Poor
200 - 300	Very poor
> 300	Unsuitable

Tab. 3 - Descriptive statistics of water analytical results.

Tab. 3 - Valori statistici che descrivono i risultati delle analisi delle acque.

Parameter	Unit	Min.	Max.	Mean	SD	WHO standard (2022)
Potential of hydrogen (pH)	/	6.70	7.97	7.44	0.26	6.5 - 8.5
Electrical conductivity (EC)	$\mu\text{S/cm}$	1993	16990	5808.41	3.35	≤ 2000
Calcium (Ca^{+2})	mg/L	45	915	215.38	156.16	≤ 75
Magnesium (Mg^{+2})	mg/L	72	645	219.78	113.71	≤ 50
Sodium (Na^+)	mg/L	187.60	2624	625.39	549.96	≤ 200
Potassium (K^+)	mg/L	2.60	94	10.04	15.23	≤ 20
Chloride (Cl^-)	mg/L	300	4400	1023.58	863.25	≤ 250
Sulfate (SO_4^{-2})	mg/L	350	3200	1103.38	662.91	≤ 250
Bicarbonate (HCO_3^-)	mg/L	64	308	195.22	57.90	≤ 120 (Algeria limit)
Nitrate (NO_3^-)	mg/L	2	332	46.73	64.03	≤ 50
Total dissolved solids (TDS)	mg/L	1190	10400	3463.78	2062.55	≤ 1000
Total hardness (TH)	mg/L as CaCO_3	479.66	3784.56	1441.83	715.14	/
Magnesium hazard (MH)	/	/	0.1644	0.7907	0.5226	0.1234
Bicarbonate hazard (BH)	mg/L	-927	20	-239.946	222.385	/
Potential salinity (PS)	mg/L	485	5500	1575.27	1089.634	/

Results and discussion

General Hydrochemistry

The descriptive statistics for the analytical results of the 37 water samples collected from various boreholes are presented in Table 3.

The pH values range from 6.7 to 7.97 (average value 7.44). These waters can be classified as neutral to slightly basic mineral waters ($6.5 \leq \text{WHO} \leq 8.5$).

The electrical conductivity (EC) values range from 1993 to 16900 $\mu\text{S/cm}$ (average value 5808.41 $\mu\text{S/cm}$), indicating that the waters of this system are highly mineralized (WHO ≤ 2000 $\mu\text{S/cm}$). Figure 4 illustrates the electrical conductivity measurements and mineralization. 100 % of total groundwater samples show very strong mineralization (WHO ≤ 1000 mg/L).

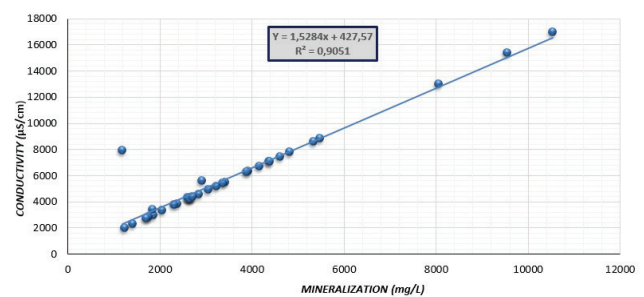


Fig. 4 - Graphical representation of electrical conductivity versus mineralization.

Fig. 4 - Rappresentazione grafica della conducibilità elettrica rispetto alla mineralizzazione.

The scatter plot reveals a clear positive correlation between total hardness (TH) and TDS (Fig. 5), indicating that increasing water mineralization is governed by the same geochemical processes, primarily the dissolution of evaporitic minerals such as gypsum and halite. Samples F86, F155, F55, F24, F17, and F142 are characterized by particularly high TH

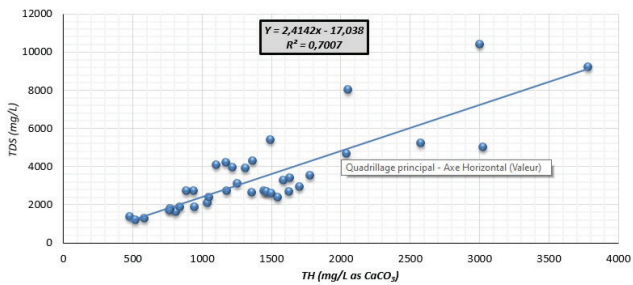


Fig. 5 - Graphical representation of total dissolved solids (TDS) versus total hardness (TH).

Fig. 5 - Rappresentazione grafica dei solidi totali disciolti (TDS) rispetto alla durezza totale (TH).

and TDS values, reflecting highly mineralized waters. This pattern suggests zones of enhanced water–rock interaction, intense mineral dissolution, and prolonged groundwater residence time within saline formations. Overall, total hardness can be considered a robust proxy for assessing the degree of groundwater salinization in the study area.

The sodium content varies from 187.6 to 2624. mg/L with an average value of 625.39 mg/L (WHO ≤ 200 mg/L). The magnesium values range from 72 to 645 mg/L (average value 219.78 mg/L) and the calcium values vary from 45 to 915mg/L (average value 215.38 mg/L). Higher electrical conductivity values are observed in areas characterized by elevated concentrations of alkali cations ($\text{Na}^+ + \text{K}^+$), particularly sodium (Na^+), indicating an increased degree of groundwater mineralization (Fig. 6).

The sulfate content varies from 350 to 3200 mg/L with an average of 1103.38 mg/L. The chloride values range from 300 to 4400 mg/L (average value. 1023.58 mg/L) and the bicarbonate values varies from 64 to 308 mg/L (average value. 195.22 mg/L). The electrical conductivity increases at points where sulfate (SO_4^{-2}) concentrations are high (WHO ≤ 250 mg/L) (Fig. 7).

Following the examination of Pearson's correlation coefficient (Table 4), strong positive correlations ($r > 0.7$) are observed between the following pairs: [Cl^- ; ($\text{Na}^+ + \text{K}^+$)] ($r = 0.99$), [TDS ; ($\text{Na}^+ + \text{K}^+$)] ($r = 0.93$), [TDS ; Cl^-] ($r = 0.93$), [TDS ; SO_4^{-2}] ($r = 0.84$), [TDS ; Mg^{+2}] ($r = 0.80$), as well as

Tab. 4 - Pearson Correlation Matrix (Pearson).

Tab. 4 - Matrice di correlazione di Pearson (Pearson).

	pH	Ca^{+2}	Mg^{+2}	$\text{Na}^+ + \text{K}^+$	HCO_3^-	Cl^-	SO_4^{-2}	NO_3^-	EC	Mineralization	TDS
pH	1.00										
Ca^{+2}	-0.28	1.00									
Mg^{+2}	-0.20	0.38	1.00								
$\text{Na}^+ + \text{K}^+$	-0.06	0.33	0.63	1.00							
HCO_3^-	-0.18	0.23	0.04	-0.17	1.00						
Cl^-	-0.13	0.35	0.64	0.99	-0.15	1.00					
SO_4^{-2}	-0.10	0.76	0.79	0.62	0.03	0.58	1.00				
NO_3^-	-0.62	-0.10	-0.01	-0.25	0.08	-0.22	-0.21	1.00			
EC	-0.12	0.52	0.76	0.91	-0.13	0.90	0.78	-0.23	1.00		
Mineralization	-0.15	0.56	0.81	0.94	-0.07	0.93	0.83	-0.20	0.96	1.00	
TDS	-0.16	0.57	0.80	0.93	-0.06	0.93	0.84	-0.20	0.96	1.00	1.00

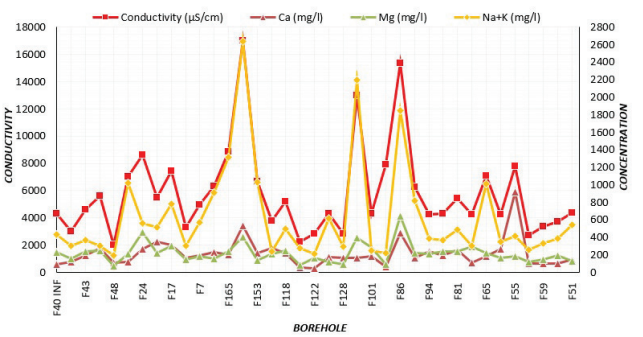


Fig. 6 - Graphical representation of electrical conductivity versus major cations.

Fig. 6 - Rappresentazione grafica della conducibilità elettrica e dei principali cationi.

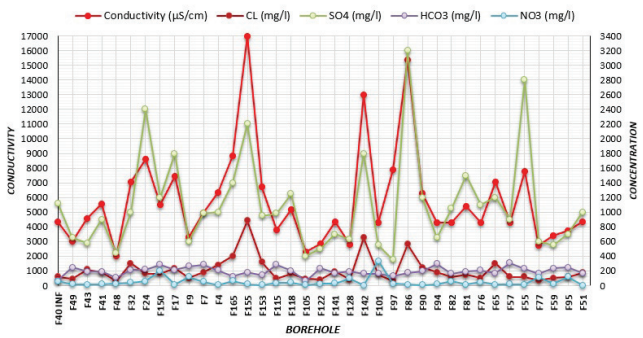


Fig. 7 - Graphical representation of electrical conductivity versus major anions.

Fig. 7 - Rappresentazione grafica della conducibilità elettrica e dei principali anioni.

[SO_4^{-2} ; Ca^{+2}] ($r = 0.76$) and [SO_4^{-2} ; Mg^{+2}] ($r = 0.77$). The high positive correlations ($r > 0.8$) between TDS and Na^+ , Cl^- , SO_4^{-2} , and Mg^{+2} indicate that these ions are major contributors to groundwater salinity. The strong correlations suggest that as the concentration of these ions increases, the TDS level also increases, and vice versa. A strong correlation between [Cl^- , ($\text{Na}^+ + \text{K}^+$)] generally indicates a salinization process, often related to halite dissolution, wastewater, and agricultural runoff. A strong correlation between [SO_4^{-2} ; Ca^{+2}] generally suggests gypsum (anhydrite) dissolution. A strong correlation between [SO_4^{-2} ; Mg^{+2}] suggests that these ions are controlled

by common geochemical processes, primarily the dissolution of evaporites associated with carbonates, ion exchange, and the effects of evaporation, indicating a dominant lithological control on the composition of groundwater (Ouda & Tidjini, 2014; Brahmi, 2015; Kabour & Chebbah, 2017; Sellami & Rabah, 2017) (Fig. 8).

The Principal Component Analysis (PCA) of the groundwater quality data from the Béchar-Kénadsa Coal Basin (Southwestern Algeria) provides a robust statistical

method to identify the dominant processes controlling hydrogeochemistry by reducing the dimensionality of the dataset. As shown in Table 5, out of a total of 13 principal components, 11 were retained in accordance with the Kaiser criterion (eigenvalues > 1).

Tab. 5 - Eigenvalues of groundwater from the Bechar-Kenadsa Coal Basin.

Tab. 5 - Valore caratteristico delle acque sotterranee del Bacino Carbonifero di Bechar-Kenadsa.

PC	Eigenvalue	% variance
1	2.07314E07	96.487
2	457425	2.1289
3	262397	1.2212
4	13450.6	0.062601
5	12140.3	0.056502
6	4177.17	0.019441
7	2976.13	0.013851
8	1528.83	0.0071154
9	465.999	0.0021688
10	214.165	0.00099675
11	111.959	0.00052107
12	0.0265563	1.236E-07
13	0.00292664	1.3621E-08

Values in bold were retained following the kaiser criterion of >1.

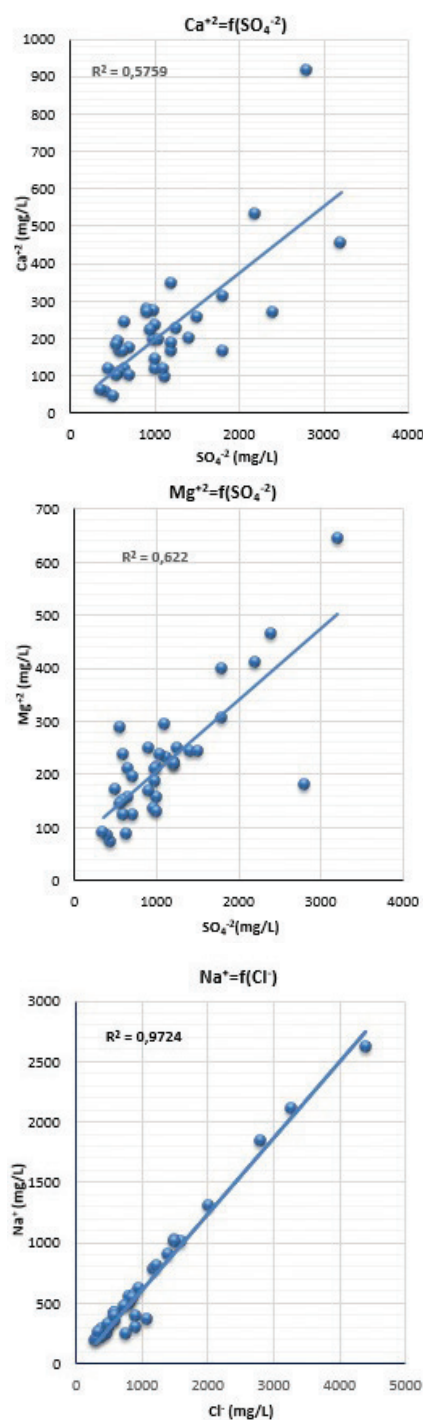


Fig. 8 - Scatter plots of hydrochemical parameters.

Fig. 8 - Grafici della dispersione dei parametri idrochimici.

The first three principal components (PCs) were overwhelmingly dominant, collectively explaining 99.84% of the total variance in the data (PC1: 96.49%, PC2: 2.13%, PC3: 1.22%) (Table 6).

PC1 (96.49% variance) was characterized by moderate positive loadings on Electrical Conductivity (EC = 0.73) and weak positive loadings on Mineralization (0.458), and TDS (0.447). This loading pattern is characteristic of bulk salinity and high mineralization. The near absence of temperature or pH influence confirms that the primary driver is not a simple dissolution reaction but rather an accumulation of dissolved solids. In the context of the Béchar-Kenadsa Basin, a coal bearing region far from the coast, this signal is not due to seawater intrusion, despite what the Gibbs diagram might falsely suggest. Instead, PC1 reflects anthropogenic enrichment from coal mining activities: leaching of coal mine tailings and combustion residues (rich in sulfates, chlorides, and trace metals) into groundwater, superimposed on naturally mineralized conditions.

PC2 (2.13% variance) was defined by moderate negative loading on EC (-0.68) and weak positive loadings on Mineralization (0.485) and TDS (0.49). The negative loading on EC coupled with positive mineralization and TDS loadings suggests a bipolar salinity gradient distinguishing high-conductivity samples from those with higher dissolved mineral content but lower conductivity. This may reflect variations in the type of dissolved solids rather than absolute concentration potentially distinguishing between carbonate-dominated versus evaporite-dominated mineralization pathways.

Tab. 6 - PCA loadings of groundwater from the Bechar-Kenadsa Coal Basin

Tab. 6 - PCA delle acque sotterranee del Bacino Carbonifero di Bechar-Kenadsa.

	PC 1	PC 2	PC 3	PC 4	PC 5	PC 6	PC 7	PC 8	PC 9	PC 10	PC 11
T (°C)	5.13E-06	-7.10E-06	1.69E-05	-5.86E-05	-6.56E-05	0.0002	-5.87E-05	-0.0008	0.0005	-0.0001	-0.0012
pH	-8.49E-06	-3.90E-05	-5.92E-06	-0.0006	-0.0014	-0.0013	0.0010	-0.00027	-0.0009	0.0014	0.0018
EC (µS/cm)	0.7293**	-0.6795**	0.0781	0.0074	0.0066	0.0080	0.0070	0.0032	0.0029	-0.0033	-0.0019
Ca ²⁺ (mg/L)	0.0186	0.0452	0.1605	0.8292***	0.0788	-0.0454	-0.0201	-0.2546	0.2249	0.3947*	0.0663
Mg ²⁺ (mg/L)	0.0196	0.0246	0.0572	-0.3588*	0.3493*	-0.207	-0.0455	0.3648*	0.4111*	0.6308**	0.0077
Na ⁺ (mg/L)	0.1128	0.0810	-0.3335*	-0.1296	-0.6503	0.4263*	0.1859	-0.1164	0.2827	0.3465*	-0.0067
K ⁺ (mg/L)	0.0016	-0.0020	-0.0082	-0.0174	-0.0260	0.0526	0.0061	0.1106	-0.1352	0.0351	0.9821***
HCO ₃ ⁻ (mg/L)	-0.0012	0.0177	0.0215	0.1138	0.2088	0.0801	0.8620***	0.2078	0.2794	-0.2675	0.0227
Cl ⁻ (mg/L)	0.1759	0.1183	-0.6031**	0.2475	-0.1106	-0.4042*	-0.2314	0.3351	0.3120*	-0.3002*	0.0354
SO ₄ ²⁻ (mg/L)	0.1182	0.2031	0.6992**	-0.0233	-0.4584*	-0.1339	-0.1273	0.2331	0.3202*	-0.2313	0.0276
NO ₃ ⁻ (mg/L)	-0.0029	0.0069	-0.0002	-0.0195	0.369*	0.6361**	-0.3760*	-0.0683	0.4680*	-0.3001*	0.0605
Mineralization	0.4581*	0.4846*	-0.0156	-0.2512	0.1782	-0.2471	0.0904	-0.6040**	0.1076	-0.0700	0.0984
TDS (mg/L)	0.4471*	0.4885*	0.0161	0.1682	0.1198	0.3422*	-0.0236	0.4440*	-0.4169	0.1219	-0.1233

*Parameters have weak loading; **Parameters have moderate loading; ***Parameters have strong loading. - T = Water temperature; EC = Electrical conductivity.

PC3 (1.22% variance) showed a complex pattern with moderate negative loadings on Cl⁻ (-0.60) and weak negative loadings on Na⁺ (-0.33), counterbalanced by a moderate positive loading on SO₄²⁻ (0.699). This inverse relationship between chloride and sulfate is a key indicator of differential evaporite mineral dissolution. The dominance of sulfate over chloride suggests that gypsum/anhydrite dissolution is more prevalent than halite dissolution in the subset of samples driving this component. The negative sodium loading further supports this, as halite dissolution would typically couple Na⁺ with Cl⁻. This pattern aligns with hydrogeochemical studies in the broader B  char region identifying sulfate-calcium waters linked to gypsiferous formations (Lachache et al., 2018; 2026).

In synthesis, the PCA robustly confirms that the groundwater chemistry in the B  char-K  nadsa Coal Basin is predominantly controlled by anthropogenic-enhanced mineralization derived from coal mining activities (tailings and

combustion residue leaching) and natural geogenic processes. The dissolution of evaporite minerals (halite, gypsum) from Upper Cretaceous formations is the overarching mechanism (PC1), driving the extreme salinization observed across the basin. This is supplemented by distinct and identifiable processes of differential evaporite dissolution (PC3: sulfate vs. chloride). The extreme variance concentration in PC1 (96.49%) reflects the homogeneous, salinity-dominated hydrochemical regime characteristic of deep continental basins in hyper-arid environments, where evaporite dissolution and evaporative concentration override all other processes (Lachache et al., 2018; 2026; Amri et al., 2023).

The Gibbs diagram suggests that water quality is strongly influenced by seawater intrusion (Fig. 9). However, given that the B  char Coal Basin is located far inland, this interpretation is most likely a false positive. The signal interpreted as seawater intrusion probably results from the highly mineralized conditions of the study area, combined with anthropogenic

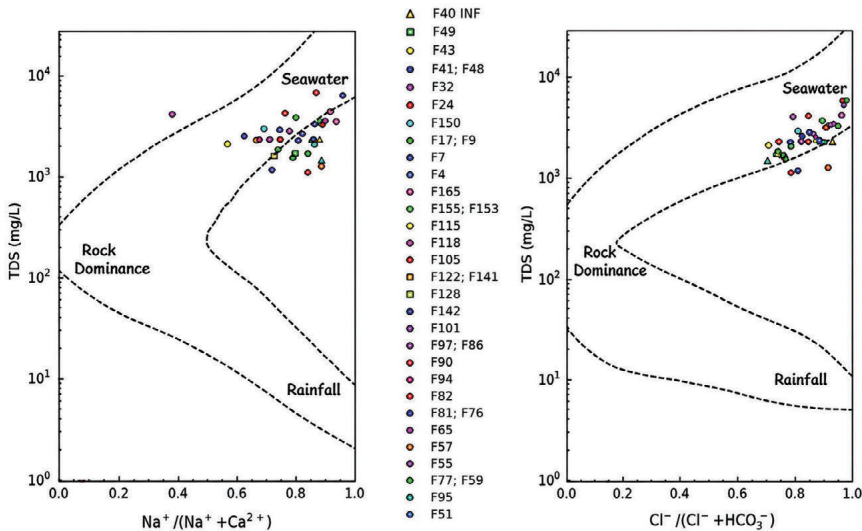


Fig. 9 - Groundwater representation on Gibbs Diagram.

Fig. 9 - Rappresentazione delle acque sotterranee attraverso il diagramma di Gibbs.

coal mining activities. These activities generate millions of tons of coal mine tailings and combustion residues, which leach into the soil and groundwater.

According to the Piper diagram (Fig. 10), the cation facies show that most of the points are clustered in the zone between calcium and magnesium, while the proportions of Na^+ and K^+ remain low. This indicates that the dominant cations are Ca^{+2} and Mg^{+2} , which is likely due to the prolonged contact of the water with carbonate rocks. Regarding the anion facies, the samples are located in the zone dominated by sulfates (SO_4^{-2}) and chlorides (Cl^-), with a minor contribution from carbonates (HCO_3^-). Therefore, the major anions are SO_4^{-2} and Cl^- . Overall, the hydrochemical facies indicate that most of the points fall within the field corresponding to calcium–magnesium waters of the chloride–sulfate type (Ca^{+2} , Mg^{+2} , Cl^- , SO_4^{-2}), while a few points are plotted in the field corresponding to sodium–potassium chloride or sodium sulfate waters which may be explained by a local anthropogenic influence, particularly related to agricultural activities (irrigation, nitrogen and potassium fertilizers) as well as domestic wastewater discharges, which can increase sodium and potassium concentrations in certain areas.

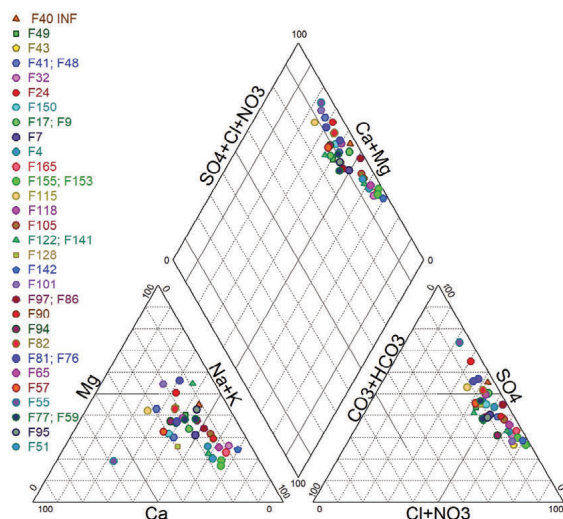


Fig. 10 - Groundwater representation on Piper's Diagram.

Fig. 10 - Rappresentazione delle acque sotterranee attraverso il diagramma di Piper.

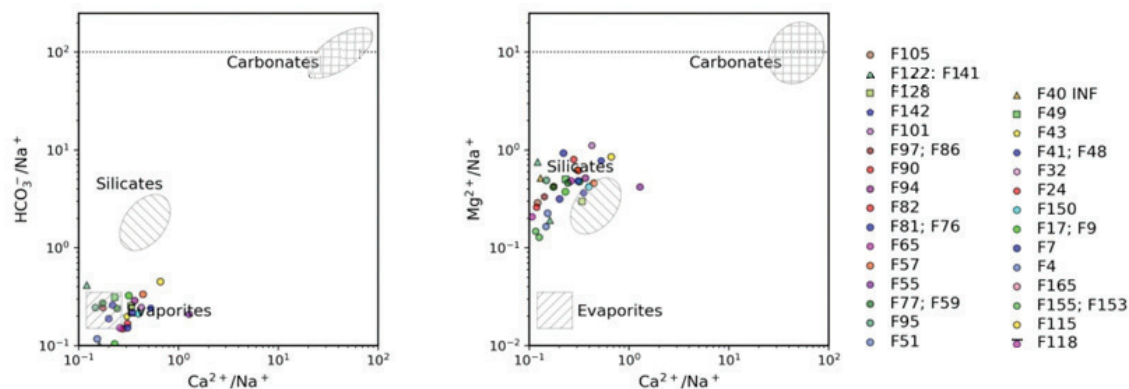


Fig. 11 - Groundwater representation on Gaillardet Diagram.

Fig. 11 - Rappresentazione delle acque sotterranee attraverso il diagramma di Gaillardet.

Thus, these waters predominantly belong to the Calcium-Magnesium Chloride-Sulfate facies, indicating a high degree of mineralization controlled by limestone dissolution, evaporite dissolution (gypsum and anhydrite), and evaporation processes in a climatic context from arid to semi-arid, as well as potential anthropogenic inputs, particularly with respect to nitrates. The relative homogeneity of the data points suggests a similar geochemical origin of the groundwater and a dominant lithological control on its chemical composition.

Gaillardet diagram shows that the origin of the water is from the weathering of evaporite and silicate rocks (Fig. 11).

The Schoeller-Berkaloff diagram shows the dominance of sodium among the cations and chloride among the anions (Fig. 12). The levels of calcium, magnesium and sulphate are close to the levels of the most dominant ions. It can be inferred that the source of sodium and chloride in the study area is from natural geology or contamination from coal mine tailings.

Groundwater suitability for drinking and irrigation

In this subsection, groundwater is assessed for human consumption based on the PWQI and for irrigation based on the electrical conductivity (EC) and the sodium adsorption ratio (SAR).

The potable water quality index (PWQI), the water quality classification according to the PWQI is presented in Table 7.

For this study, the results indicate that water is not suitable for human consumption.

Sodium adsorption ratio (SAR) results exhibit high to very high salinity of the groundwater in the study area with a medium SAR (Fig. 13). Therefore, it is safe from sodium hazard but unsuitable for irrigation due to a high risk of soil salinization. The salinity can lead to significant harmful effects due to the fixation of sodium from sodium chloride salts by soil colloids. The Sodium has a harmful effect on vegetation, as it degrades the physical properties of the soil, making it compact and suffocating for plants (Todd, 1980). When the concentration of soluble Na^+ ions in the soil is high, these ions often replace Ca^{+2} cations in the exchange complex. Water with high salt content can cause this effect. The risk is determined from the value of absorbable sodium, the Sodium

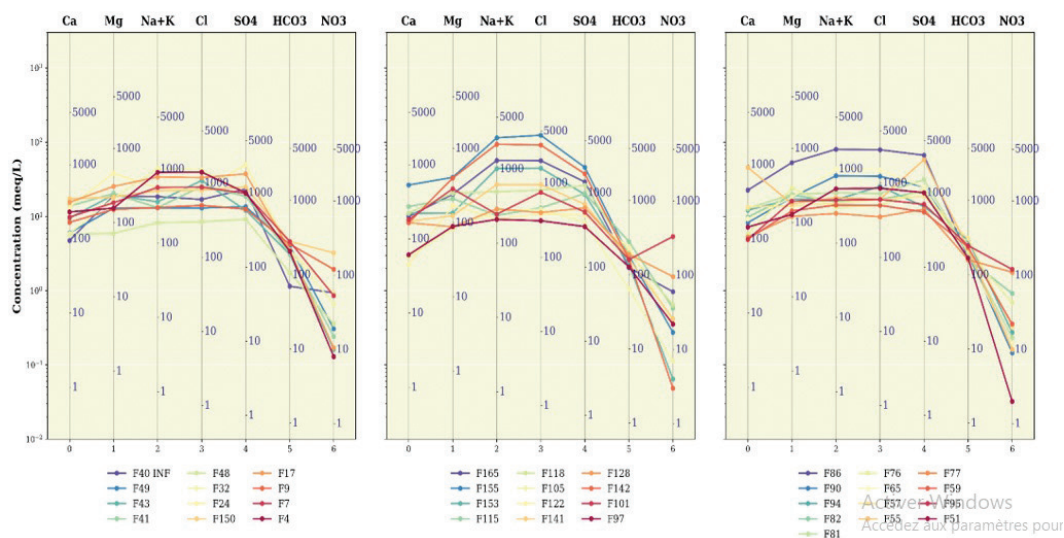


Fig. 12 - Groundwater representation on the Schoeller - Berkaloff Diagram.

Fig. 12 - Rappresentazione delle acque sotterranee attraverso il diagramma di Schoeller - Berkaloff.

Tab. 7 - Water class based on PWQI values (Aly et al., 2015).

Tab. 7 - Classificazione delle acque basata su i valori PWQI (Aly et al., 2015).

PWQI	Water class	Samples
< 50	Excellent	No samples
50 - 100	Passable	No samples
100 - 200	Poor	F49, F48, F105, F122, F97, F59
200 - 300	Very poor	F40 INF, F43, F41, F9, F7, F153, F115, F118, F141, F128, F94, F82, F81, F76, F57, F77, F95, F51
> 300	Unsuitable	F32, F24, F150, F17, F4, F165, F155, F142, F101, F86, F90, F65, F55

Adsorption Ratio (SAR). For the same electrical conductivity, the risk increases as the SAR value becomes higher (Amri et al., 2023).

Pollution Index Results; According to World Health Organization (WHO) standards, nitrate concentrations exceeding 50 mg/L are considered a primary indicator of groundwater pollution. Analysis of the study area identified several contaminated boreholes: F40 INF, F24, F150, F9, F7, F165, F128, F101, F82, F77, and F95. All other boreholes in the study area were found to be within safe limits and are not classified as contaminated (Fig. 14).

The exceptionally high nitrate (NO_3^-) concentrations observed in samples F101 (332 mg/L) and F150 (200 mg/L) are attributed to localized anthropogenic pollution sources originating from the surface, as both sampling points are located in areas with intensive agricultural activity. The use of nitrogen-based fertilizers, such as ammonium nitrate and urea, in cultivated lands leads to nitrate leaching toward the groundwater through the unsaturated zone, contributing to groundwater contamination.

The consumption of water from the eleven contaminated boreholes identified (F40 INF, F24, F150, F9, F7, F165, F128, F101, F82, F77, and F95) poses significant health risks to local populations. It is critical to note that, according to

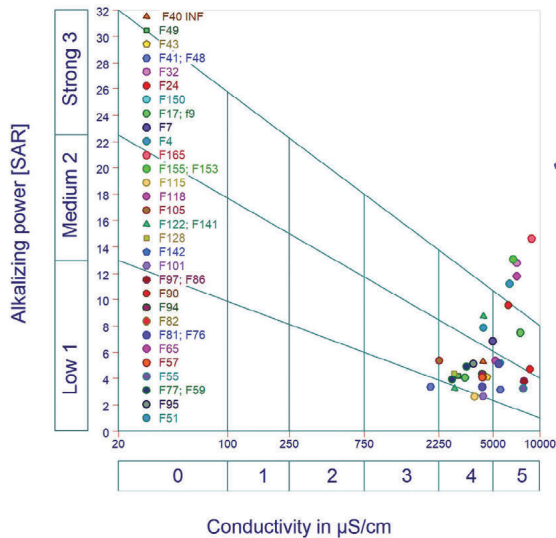


Fig. 13 - Groundwater representation on SAR Diagram.

Fig. 13 - Rappresentazione delle acque sotterranee attraverso il diagramma di SAR.

World Health Organization (WHO) guidelines, nitrate concentrations exceeding 50 mg/L are particularly hazardous for vulnerable groups. High nitrate intake is associated with serious conditions such as methemoglobinemia (commonly known as 'blue baby syndrome') in infants, and it also presents substantial risks for pregnant women.

The analyses of groundwater in the study area reveal a high degree of mineralization. The EC values range from 1993 $\mu\text{S}/\text{cm}$ to 16990 $\mu\text{S}/\text{cm}$, significantly exceeding the WHO guideline of 2000 $\mu\text{S}/\text{cm}$, which indicates that the water is highly mineralized. These waters are not suitable for drinking, as they exhibit high concentrations of major elements. Sodium levels are very high, ranging from 187,6 mg/L to 2624 mg/L (WHO ≤ 200 mg/L). Magnesium concentrations are also elevated, varying from 72 mg/L to 645 mg/L (WHO ≤ 50 mg/L). Sulfate levels range from 350 mg/L to 3200 mg/L (WHO ≤ 250 mg/L), while Chloride concentrations vary between 300 mg/L and 4400 mg/L (WHO ≤ 250 mg/L). Calcium shows high concentrations from

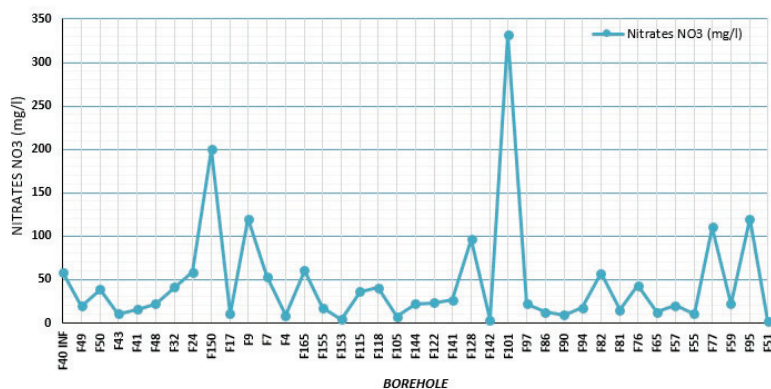


Fig. 14 - Nitrate concentrations in the groundwater of the "Coal Basin".

Fig. 14 - Concentrazione di nitrati nelle acque sotterranee di "Coal Basin".

45 mg/L to 915 mg/L (WHO ≤ 75 mg/L), and Potassium ranges from 2.6 mg/L to 94 mg/L (WHO ≤ 20 mg/L). Nitrate concentrations are also elevated at certain points, indicating localized pollution. All these values exceed both the WHO guidelines for drinking water quality (2022). This situation is influenced by several interrelated factors. The sandstone beds of the aquifer are exposed at the surface in the study area and are affected by fracturing. The Westphalian aquifer is fragile and corresponds to an unconfined groundwater system, making it particularly vulnerable to both natural and anthropogenic influences. High concentrations of cations and anions are primarily related to the lithological characteristics of the aquifer, particularly the presence of evaporitic rocks such as gypsum and halite. These formations enhance salt dissolution and increase ion concentrations, leading to higher groundwater electrical conductivity. In addition, evaporation processes, especially in arid and semi-arid environments, further contribute to salinity increase, thereby leading to higher electrical conductivity of water. The Upper Westphalian formation contributes to the observed high mineralization due to prolonged water-rock interactions along groundwater flow paths. Coal genesis in this region occurs through a complex geological process in lagoon and Sebkha environments, controlled by climatic conditions, eustatic sea-level changes, subsidence, and temperature. The sedimentary sequences, known as cyclothems (Deleau, 1951; 1952; 1955), include sulfate-bearing and evaporitic rocks, which influence both the hydrogeological behavior of the aquifer and the chemical composition of groundwater. Coal extraction in the Bechar mining basin has generated numerous solid residues stored in the open air as spoil heaps. These spoil heaps are exposed to weathering, and the materials composing them are altered by meteorological factors. These stored spoil heaps are mainly composed of coal-bearing shales and contain metallic sulfides, predominantly pyrite, which can be oxidized during infiltration and percolation of rainwater. The resulting sulfates and metals may migrate and reach the underlying aquifer, contributing to groundwater mineralization and potentially affecting water quality. In addition, uncontrolled discharges, such as urban wastewater, are continuously increasing, significantly exacerbating the deterioration of water quality and raising nitrate concentrations in groundwater. Excessive use of fertilizers, combined with uncontrolled irrigation and

over-application of mineral fertilizers, also negatively impacts groundwater quality, primarily causing diffuse nitrate pollution ($\text{NO}_3^- > 50$ mg/L) (Amri et al., 2023; Belkendil et al., 2018; 2021; 2025; Sellami & Rabah, 2017). Identifying the causes of pollution, particularly those stemming from human activities, is a key step in developing sustainable solutions that protect groundwater quality and support sustainable development.

Conclusion

The hydrogeological and hydrochemical study of groundwater in the Kenadsa coal basin has revealed that the Westphalian aquifer (Upper Carboniferous) consisted of a complex lithological assemblage dominated by sandstones. The groundwater in the study area exhibits high mineralization, with total dissolved solids (TDS) values ranging from 1176 mg/L to 10534 mg/L. Interpretation of the hydrochemical plots, notably the Piper and Schoeller diagrams, reveals that the dominant hydrochemical facies are calcium-magnesium chlorides and sulfates, reflecting strong water-rock interactions and possible anthropogenic inputs. The pH values from 6.70 to 7.97 indicate slightly acidic to neutral conditions, while the electrical conductivity from 1993 $\mu\text{S}/\text{cm}$ to 16990 $\mu\text{S}/\text{cm}$, confirms a high degree of mineralization. Major cation concentrations are notably high, with calcium (Ca^{+2}) concentrations ranging from 45 to 915 mg/L, and magnesium (Mg^{+2}) from 72 to 645 mg/L. The elevated concentrations of other major ions further demonstrate significant chemical enrichment, rendering the groundwater unsuitable for drinking according to WHO guidelines for drinking water quality. Despite this, the SAR diagram indicates that the groundwater remains suitable for irrigation. Nitrate concentrations exceed the WHO in several boreholes. This nitrate enrichment suggests additional anthropogenic pressures on the aquifer.

The observed degradation of water quality is likely the result of multiple, overlapping factors:

the lithological composition of the Westphalian aquifer, which favors intense water-rock interactions; uncontrolled discharge of untreated wastewater into the environment; the high intrinsic vulnerability of the unconfined Westphalian aquifer; excessive use of agricultural fertilizers; and the dissolution and leaching of spoil heaps generated by coal

mining activities in the region. Groundwater protection depends on wastewater treatment, reduced fertilizer use, protection of capture zones, regular water quality monitoring, and sustainable management.

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Author contributions

The study was carried out and written by the first author Hadji ayyoub under the supervision and guidance of all co-authors (Mekkaoui A., Habi M., Belkendil A. and Amri S.). All authors contributed to reviewing, revising, and approving the final manuscript.

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Competing interest

The authors declare no competing interest.

Supplementary Materials

The following supporting information can be downloaded at: www.acquesotterranee.net/acque/article/view/983

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